Satellite Imagery Archaeological Analysis

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The primary aim of this web application is to assist archaeologists, researchers, and remote sensing professionals in analyzing satellite imagery to detect and classify potential archaeological and geographical features.

Objectives:
- Automate the preliminary visual inspection of satellite imagery using advanced artificial intelligence.
- Provide a standardized classification system for identifying settlements, agricultural modifications, soil features, water management systems, and defensive structures.
- Enable temporal and comparative analysis to detect structural changes, excavations, or land-use changes over time.
- Offer an interactive, intuitive workspace for experts to annotate, zoom, and inspect high-resolution imagery.
- Streamline the reporting process by exporting automated findings into shareable PDF documents.

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- **Core Programming Language:** The application is built entirely using **TypeScript** (a statically typed superset of **JavaScript**). Python was *not* used for the development of this web application.
- **Frontend Framework:** React 19, initialized via Vite, utilizing Tailwind CSS for responsive and modern UI styling.
- **Image Manipulation & UI:** `react-zoom-pan-pinch` for the interactive canvas workspace, `lucide-react` for iconography.
- **Reporting:** `jspdf` and `html2canvas` are utilized to capture the workspace and generate downloadable PDF reports.

ï¿½>ï¿½ï¿½ M a c h i n e L e a r n i n g & A I I n t e g r a t i o n
Rather than training a custom, specialized machine learning model from scratch using Python (e.g., PyTorch or TensorFlow), this application leverages **Google's Gemini Multimodal Large Language Model (LLM)**.

Specifically, it uses the `@google/genai` SDK to communicate with the

Gemini Flash model.
- **Methodology:** The AI acts as an expert system. It is supplied with a strict "System Instruction" containing an extensive classification key (covering climate zones, LULC, burial funerary types, etc.).
- **Computer Vision:** The multimodal capabilities of Gemini allow it to "see" the uploaded satellite imagery (encoded as Base64) and cross-reference visual anomalies against the provided archaeological key.
- **Structured Output:** The application uses strict JSON Schemas to force the AI to return exact coordinate bounding boxes (`x, y, width, height`) for automatic feature detection, alongside generating descriptive Markdown reports.

ï¿½=Þ€ What the WebApp Can Do (Features)
1. **Single Image Analysis:** Upload a satellite image and run an AI scan to generate a comprehensive archaeological report.
2. **Auto-Detect Features:** The AI scans the image and automatically draws bounding boxes over areas of interest (e.g., "Burial Mound", "Embankment").
3. **Compare Analysis (A/B Testing):** Upload two images (e.g., historical vs. current) to visually highlight and document differences, such as new excavations or deforestation.
4. **Targeted / Manual Annotation:** Users can manually draw bounding boxes or annotations on the image to force the AI to focus its analysis *only* on the highlighted regions.
5. **Batch Analysis:** Queue up multiple satellite images for sequential AI processing.
6. **History & PDF Export:** All analyses are saved in a local History log. Users can export their annotated images and AI-generated markdown reports directly to a PDF file for documentation and sharing.

&ï¿½ï¿½ How It Functions (User Workflow)
1. **Upload:** The user selects the desired module (Single, Compare, or Batch) from the sidebar and uploads satellite imagery (JPEG/PNG/WebP).
2. **Inspect & Annotate:** The image loads into the interactive workspace. The user can zoom, pan, and optionally draw manual annotations using the toolbar.
3. **Analyze:** The user clicks "Run AI Scan". The image is converted

to a secure format and sent to the Gemini AI API along with the system instructions and any user-provided context notes.
4. **Review:** The AI returns structured coordinates and a detailed report. The workspace automatically draws the detected features onto the image canvas.
5. **Export:** The user reviews the findings, adjusts as needed, and clicks "Export to PDF" to generate a final, shareable document combining the annotated imagery and the text report.